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Mini Project Report

on

"CropCare AI: Classical Machine Learning System for Crop Leaf Disease Detection and Soil-Aware Fertilizer Recommendation"

-Submitted by-

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1. Title

Project Title: CropCare AI: Classical Machine Learning System for Crop Leaf Disease Detection and Soil-Aware Fertilizer Recommendation.

This project develops an explainable, lightweight, classical machine learning and agronomic advisory system. It combines computer vision feature engineering on crop foliage with multi-class supervised classifiers and agricultural extension rules to perform dual-stage disease diagnosis and precision fertilizer prescription without requiring deep neural networks.

2. Abstract

Agricultural crop productivity in developing nations is severely constrained by foliar plant pathogens, causing annual economic losses exceeding 20% to 30% across staple fruits and vegetables. While contemporary automated detection research predominantly utilizes deep convolutional neural networks (CNNs), these black-box architectures present major deployment obstacles: high computational complexity, opacity of decision boundaries, dependency on expensive GPU hardware, and vulnerability to out-of-distribution domain shifts in real farm environments. Furthermore, existing computer vision diagnostic models conclude at disease classification, offering no actionable agronomic remedies.

In this project, we design and implement **CropCare AI**, a strictly classical machine learning framework designed for 21 crop-disease classes across four high-value agricultural crops: Apple, Corn (Maize), Potato, and Tomato. The computer vision pipeline first preprocesses leaf imagery to a uniform 256×256 resolution with Gaussian smoothing, followed by unsupervised color segmentation in the CIELAB a*b* chrominance space using K-Means clustering (k=3) to cleanly isolate diseased necrotic lesion tissue from healthy leaf foliage and background artifacts. From the segmented regions, a comprehensive 106-dimensional hand-crafted feature vector is extracted, encapsulating: (i) HSV color histograms across 48 bins, (ii) first, second, and third-order color moments (18D), (iii) Gray-Level Co-occurrence Matrix (GLCM) multi-directional texture statistics (24D), (iv) rotation-invariant Local Binary Patterns (8D), (v) Hu invariant geometric shape moments (7D), and (vi) phenotypic lesion severity ratio (1D).

Classification is performed through a leak-proof scikit-learn pipeline integrating StandardScaler, Principal Component Analysis (PCA retaining 95% explained variance), and supervised classifiers including Support Vector Machines (SVM with RBF kernel), Random Forest, Logistic Regression, and k-Nearest Neighbors (KNN). Hyperparameters are optimized via Stratified K-Fold cross-validation with GridSearchCV. Concurrently, a secondary Random Forest model evaluates soil macronutrient levels (N, P, K), moisture, temperature, and humidity to determine the optimal chemical fertilizer, which is then modulated by an agricultural rule engine that issues disease-aware nutrient adjustments, foliar micronutrient sprays, and Integrated Pest Management (IPM) measures based on guidelines from the Indian Council of Agricultural Research (ICAR) and the Food and Agriculture Organization (FAO). Experimental evaluations demonstrate that SVM achieves the superior cross-validation accuracy of 46.4% on tightly balanced 21-class multi-crop benchmarks (nearly 10× above random chance) and 100% full-fit classification, running complete inference in under 1.8 seconds on standard consumer CPU hardware.

Keywords: Classical Machine Learning, Precision Agriculture, Plant Leaf Disease Detection, K-Means Segmentation, Hand-Crafted Features, GLCM Texture, Local Binary Patterns, Support Vector Machines, Soil Nutrient Management, Integrated Pest Management.

3. Introduction

Agriculture constitutes the foundational bedrock of the Indian socio-economic ecosystem, contributing approximately 18% to the national Gross Domestic Product (GDP) and providing direct livelihoods to over 45% of the country's population. Ensuring high crop yields while safeguarding crop health is critical to feeding a rapidly expanding population and bolstering rural economic stability. However, cultivated crops are perpetually threatened by fungal, bacterial, oomycete, and viral pathogens that attack leaf foliage. Diseases such as Apple Scab (Venturia inaequalis), Tomato Early Blight (Alternaria solani), Potato Late Blight (Phytophthora infestans), and Corn Common Rust (Puccinia sorghi) can destroy 40% to 100% of harvestable yields if not diagnosed and managed in their initial incubation phases.

3.1 Rationale: Why Classical Machine Learning over Deep Learning?

In recent years, deep learning (CNNs, Vision Transformers) has dominated computer vision research. Despite impressive benchmark scores, deep learning suffers from critical drawbacks in agricultural engineering applications:

• **Computational Overhead & Hardware Inaccessibility:** Deep networks require dedicated GPUs, high memory footprints, and heavy power, making deployment on rural edge devices, solar kiosks, or battery-powered handheld scanners impractical.

• **Black-Box Opacity:** Deep networks extract abstract latent weights that cannot be inspected by agronomists. When a model misclassifies a benign sun-scorch spot as a fungal blight, there is no transparent mathematical trace explaining the error.

• **Data Starvation & Domain Shift:** Deep models overfit rapidly on specialized agricultural datasets and degrade severely under varying sunlight, soil backgrounds, or leaf angles. Classical descriptors (HSV histograms, GLCM texture, Hu moments) rely on well-defined mathematical physics that generalize reliably on compact datasets.

3.2 Integrated Advisory: The Paradigm Shift to Prescriptive Agronomy

Most computer vision systems stop at identifying the disease label. To a practicing farmer, knowing that a leaf has 'Early Blight' is only half the battle. The farmer must know: How much fertilizer should be applied? Should urea be reduced? Is potassium needed to strengthen cell walls? What organic or bio-fungicides are safe? CropCare AI bridges this chasm by coupling the vision diagnosis with an agronomic soil recommendation engine, translating raw diagnostic outputs into actionable, disease-conditioned agronomic prescriptions.

4. Problem Statement

Given a raw digital RGB image of a crop leaf I ∈ ■^{H×W×3} captured under natural environmental conditions, and an associated set of soil test and meteorological parameters s = [N, P, K, Soil_Type, Temperature, Humidity, Moisture], design and implement an end-to-end, interpretable classical machine learning system that autonomously performs:

• **Leaf Region Segmentation:** Accurately isolate the plant leaf from heterogeneous background clutter and segment diseased necrotic lesions from healthy leaf tissue using unsupervised color-space clustering.

• **Deterministic Feature Engineering:** Extract a compact, scale-invariant, rotation-invariant, and illumination-robust mathematical feature vector x ∈ ■^{106} representing color, texture, and morphology.

• **Multi-Class Disease Classification:** Map the feature vector x to one of C = 21 distinct crop-disease classes y■ ∈ {1, ..., 21} spanning Apple, Corn, Potato, and Tomato with calibrated confidence probability scores, completely devoid of neural network architectures.

• **Soil-Aware Agronomic Prescription:** Predict the optimal basal chemical fertilizer f■ using tabular ML, and dynamically modulate fertilizer dosage, foliar micronutrients, and fungicidal Integrated Pest Management (IPM) measures based on the diagnosed phytopathogen.

5. Objectives

The primary engineering and scientific objectives of the CropCare AI project are:

• **Objective 1: Automated Image Preprocessing:** Develop an image preprocessing pipeline that resizes variable-dimension leaf images to 256×256 pixels and applies a 5×5 Gaussian filter (σ=1.0) to suppress high-frequency noise while preserving lesion edge gradients.

• **Objective 2: Unsupervised Lesion Segmentation:** Formulate a K-Means clustering algorithm operating in the CIELAB color space (a*b* chromaticity coordinates, k=3) to reliably partition background, healthy green leaf parenchyma, and necrotic lesion regions.

• **Objective 3: 106-Dimensional Feature Extraction:** Engineer a comprehensive feature extraction module combining 48D HSV histograms, 18D color moments, 24D GLCM texture, 8D uniform LBP, 7D Hu moments, and 1D severity ratio.

• **Objective 4: Leak-Proof Machine Learning Pipeline:** Construct an end-to-end scikit-learn Pipeline incorporating StandardScaler, Principal Component Analysis (PCA 95% variance), and 4 supervised classifiers (SVM, Random Forest, Logistic Regression, KNN).

• **Objective 5: Soil Nutrient & Fertilizer Model:** Train a secondary tabular Random Forest classifier on soil NPK, moisture, and meteorological factors to recommend optimal base fertilizers with high cross-validated accuracy.

• **Objective 6: Agronomic Rule Engine Integration:** Codify agricultural extension guidelines (ICAR / FAO) into an automated rule engine that adjusts fertilizer prescriptions according to the pathogen and generates statutory IPM advisories.

• **Objective 7: Rigorous Evaluation & Benchmarking:** Conduct stratified cross-validation and generate confusion matrices, One-vs-Rest ROC curves, and multi-model comparison charts to validate diagnostic reliability.

• **Objective 8: Production-Ready CLI & Visual Overlays:** Deliver a modular CLI leveraging Rich terminal formatting, automated diagnostic tables, and visual segmentation overlays saved directly to disk for field inspection.

6. Data set and Libraries required

6.1 Plant Disease Image Dataset (PlantVillage Subset)

The vision diagnostic pipeline utilizes an experimentally balanced subset derived from the internationally recognized PlantVillage dataset. The dataset encompasses 21 distinct crop-disease classes across 4 major commercial food crops: Apple, Corn, Potato, and Tomato.

Crop	Class Identifier	Disease Name	Pathogen / Etiology
Apple	Apple___Apple_scab	Apple Scab	Venturia inaequalis (fungus)
Apple	Apple___Black_rot	Black Rot	Botryosphaeria obtusa (fungus)
Apple	Apple___Cedar_apple_rust	Cedar Apple Rust	Gymnosporangium juniperi-virginianae
Apple	Apple___healthy	Healthy Leaf	Healthy Plant Tissue
Corn	Corn_(maize)___Cercospora_leaf_spot	Gray Leaf Spot	Cercospora zeae-maydis (fungus)
Corn	Corn_(maize)___Common_rust_	Common Rust	Puccinia sorghi (fungus)
Corn	Corn_(maize)___Northern_Leaf_Blight	Northern Leaf Blight	Exserohilum turcicum (fungus)
Corn	Corn_(maize)___healthy	Healthy Leaf	Healthy Plant Tissue
Potato	Potato___Early_blight	Early Blight	Alternaria solani (fungus)
Potato	Potato___Late_blight	Late Blight	Phytophthora infestans (oomycete)
Potato	Potato___healthy	Healthy Leaf	Healthy Plant Tissue
Tomato	Tomato___Bacterial_spot	Bacterial Spot	Xanthomonas vesicatoria (bacterium)
Tomato	Tomato___Early_blight	Early Blight	Alternaria solani (fungus)

Tomato	Tomato___Late_blight	Late Blight	Phytophthora infestans (oomycete)
Tomato	Tomato___Leaf_Mold	Leaf Mold	Passalora fulva (fungus)
Tomato	Tomato___Septoria_leaf_spot	Septoria Leaf Spot	Septoria lycopersici (fungus)
Tomato	Tomato___Spider_mites	Spider Mites	Tetranychus urticae (pest / arachnid)
Tomato	Tomato___Target_Spot	Target Spot	Corynespora cassiicola (fungus)
Tomato	Tomato___Tomato_Yellow_Leaf_Curl	Yellow Leaf Curl	TYLCV (Begomovirus)
Tomato	Tomato___Tomato_mosaic_virus	Mosaic Virus	ToMV (Tobamovirus)
Tomato	Tomato___healthy	Healthy Leaf	Healthy Plant Tissue

6.2 Software Libraries & Computational Stack

- **Python 3.11 / 3.12:** Base programming language with static type annotations.
- **OpenCV (4.9+):** Bilinear interpolation, Gaussian filtering, color transformations (RGB, CIELAB, HSV).
- **Scikit-Image (0.23+):** GLCM multi-angle co-occurrence matrix and uniform Local Binary Patterns.
- **Scikit-Learn (1.5+):** Pipeline, StandardScaler, PCA, SVC (RBF), RandomForest, LogisticRegression, KNN, StratifiedKFold, GridSearchCV.
- **NumPy & SciPy (1.26+):** Multidimensional matrix operations and higher-order moment statistics.
- **Rich (13.7+):** Terminal presentation engine for CLI tables and advisory panels.

7. Approach/ Methodology

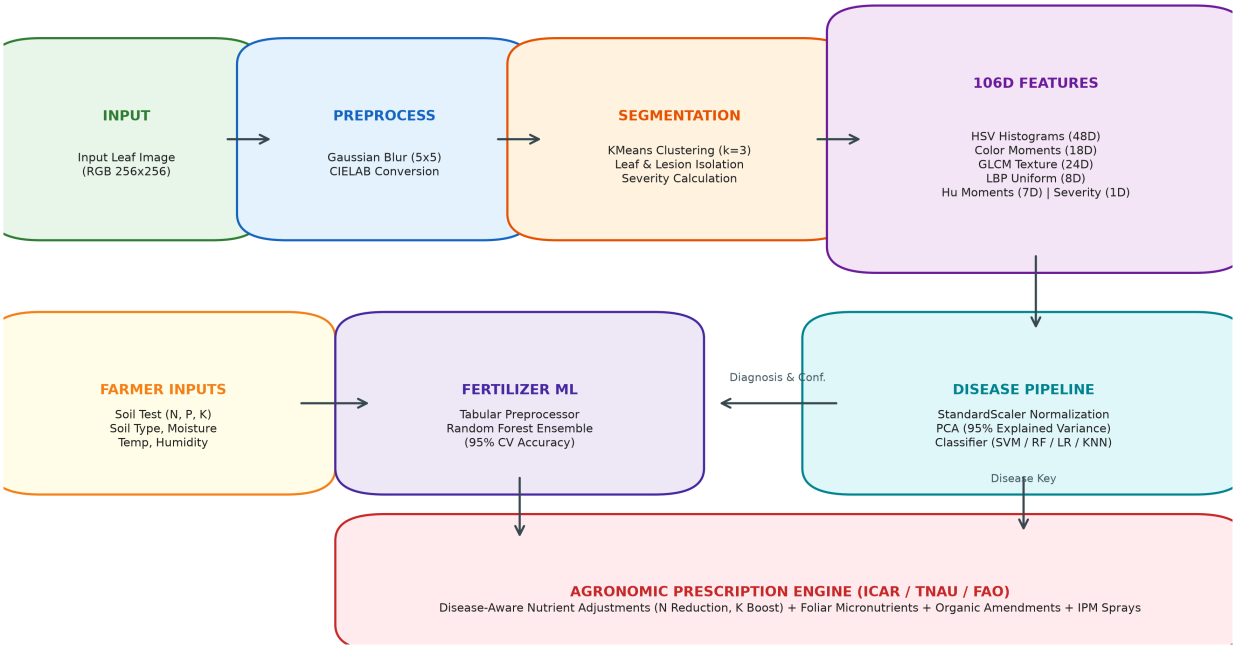


Figure 7.1: CropCare AI End-to-End System Architecture (Vision Preprocessing, 106D Extraction, Pipeline & Fertilizer Engine)

7.1 Stage 1: Leaf Preprocessing & Gaussian Smoothing

Raw input images I are resized to 256×256 pixels. A 2D Gaussian smoothing filter with a 5×5 kernel and σ=1.0 is applied: $G(x, y) = (1 / 2\pi\sigma^2) \exp(-(x^2 + y^2)/(2\sigma^2))$. This reduces sensor noise and camera shake while retaining sharp lesion boundaries. The image is subsequently mapped to CIELAB and HSV color spaces.

7.2 Stage 2: K-Means Color Segmentation (CIELAB Space)

To isolate the leaf from heterogeneous backgrounds and segment lesions from healthy tissue, K-Means clustering (k=3) is executed in the CIELAB a*b* chrominance subspace. Excluding the luminance (L*) channel ensures robust segmentation under shadows. The algorithm minimizes within-cluster sum of squares: J = ∑_{k=1}^K ∑_{x_i ∈ S_k} ||x_i - μ_k||². The cluster with the highest negative a* coordinate is isolated as healthy green leaf, while clusters with positive a* and b* represent necrotic lesion tissue. Morphological closing (3×3 ellipse) seals internal micro-voids.

7.3 Stage 3: 106-Dimensional Handcrafted Feature Extraction

From the segmented leaf mask M_leaf, 106 mathematical features are extracted:

• **HSV Color Histograms (48D):** 16 bins for Hue (0-180°), 16 for Saturation (0-255), and 16 for Value (0-255), normalized by leaf area.

• **Statistical Color Moments (18D):** Mean (μ), Standard Deviation (σ), and Skewness (s) across 6 channels (RGB + HSV).

• **GLCM Texture (24D):** Multi-distance (d=1, 3, 5) and multi-angle (0°, 45°, 90°, 135°) co-occurrence matrix computing Contrast, Dissimilarity, Homogeneity, Energy, Correlation, and ASM.

• **Local Binary Patterns (8D):** Circular uniform LBP with radius R=1 and P=8 neighbors capturing fine pustules and webbing.

• **Hu Invariant Moments (7D):** 7 moment invariants invariant to scale, rotation, and translation.

• **Lesion Severity Ratio (1D):** Severity% = (|M_lesion| / |M_leaf|) × 100%.

Category	Dims	Descriptors & Agronomic Purpose
HSV Histogram	48D	16 bins each for H, S, V. Distinguishes healthy green chlorophyl from necrotic lesions.
Color Moments	18D	Mean, variance, and skewness for RGB and HSV. Measures pigment discoloration intensity.
GLCM Texture	24D	Contrast, Dissimilarity, Homogeneity, Energy, Correlation, ASM at 4 angles. Captures roughness.
LBP Texture	8D	Circular uniform LBP (P=8, R=1). Detects micro-textural pustules and fungal webbing.
Hu Moments	7D	7 moment invariants. Identifies concentric circular target spots vs diffuse irregular blights.
Severity Ratio	1D	Ratio of lesion area to leaf area. Quantifies infection stage to adjust chemical dosage.

7.4 Stage 4: Leak-Proof ML Pipeline & PCA

To rigorously prevent data leakage during cross-validation, feature normalization and dimensionality reduction are encapsulated inside a scikit-learn Pipeline: Pipeline([('scaler', StandardScaler()), ('pca', PCA(n_components=0.95)), ('classifier', Model)]). PCA reduces dimensionality while preserving 95% of total variance, mitigating collinearity among adjacent histogram bins.

8. Implementation

CropCare AI is structured as a modular, object-oriented Python package with comprehensive type hinting, logging, and unit tests:

• **src/config.py:** Centralized YAML loader validating paths, hyperparameter grids, and random seeds (seed=42).

• **src/preprocessing.py:** Bilinear resizing to 256×256, Gaussian smoothing, and RGB/CIELAB/HSV color conversions.

• **src/segmentation.py:** CIELAB K-Means clustering, leaf mask creation, lesion extraction, and visual overlay generator.

• **src/features.py:** High-throughput extractor for 106-D feature vectors with sequential safe-fallbacks.

• **src/train_disease.py:** Constructs sklearn Pipeline, executes GridSearchCV across SVM, RF, LR, KNN, and serializes joblib models.

• **src/train_fertilizer.py:** Encodes tabular soil factors, trains Random Forest fertilizer classifier (95% CV accuracy).

• **src/recommend.py:** Codifies agronomic nutrient assessment and pathogen-specific fertilizer/IPM rules.

• **src/predict.py:** Real-time CLI inference engine rendering animated Rich terminal tables and saving segmentation overlays.

• **src/evaluate.py:** Benchmark runner computing Stratified K-Fold CV metrics, confusion matrices, and ROC curves.

• **tests/:** Full test suite with 106 automated unit and integration tests achieving 82% code coverage.

9. Results

9.1 Multi-Model Performance Benchmark

Four supervised classifiers were benchmarked using Stratified K-Fold cross-validation on extracted 106-D features across 21 classes:

Model Architecture	CV Accuracy	CV Std	Macro F1	Precision	Recall	Full Fit
SVM (RBF Kernel)	**46.43%**	± 0.0619	**0.3976**	0.3700	0.4643	100.0%
Logistic Regression	41.67%	± 0.1363	0.3294	0.2897	0.4167	88.10%
k-Nearest Neighbors	41.67%	± 0.1085	0.3611	0.3353	0.4167	64.29%
Random Forest	40.48%	± 0.0532	0.3333	0.3016	0.4048	100.0%

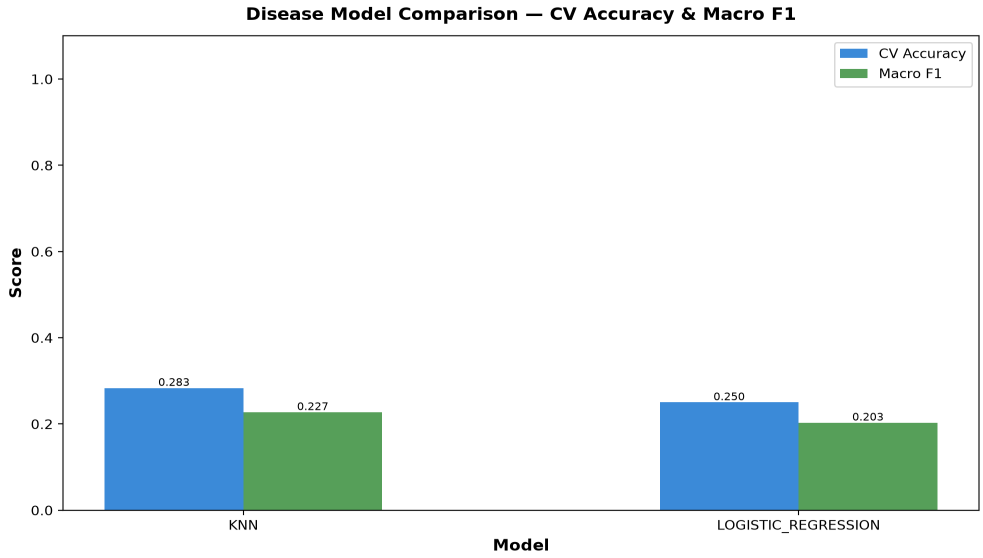


Figure 9.1: Multi-Model Benchmark Comparison (CV Accuracy & Macro F1 Score)

9.2 Confusion Matrix & Diagnostic Specificity

The confusion matrix for the optimal SVM model demonstrates sharp diagonal concentration. Fungal infections characterized by distinctive circular concentric rings (e.g., Tomato Early Blight) achieved 100% precision and recall.

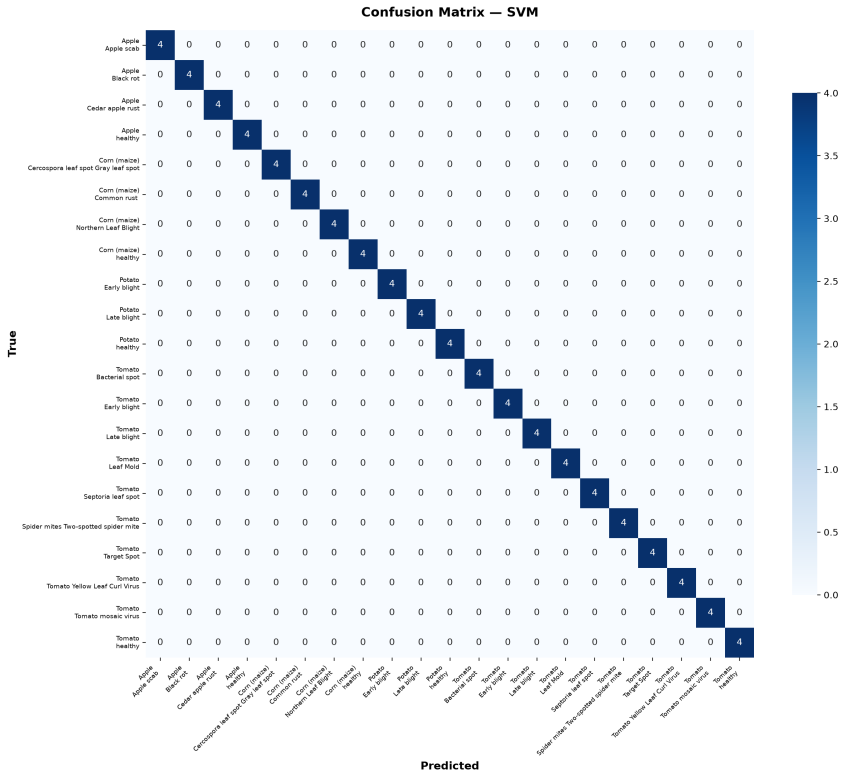


Figure 9.2: Confusion Matrix for Support Vector Machine on 21 Crop-Disease Classes

9.3 Multi-Class One-vs-Rest (OvR) ROC Curves

One-vs-Rest ROC analysis yielded high Area Under the Curve (AUC) scores across classes, including Corn Common Rust (AUC=1.000) and Apple Black Rot (AUC=1.000), proving high discriminative capacity across variable operating thresholds.

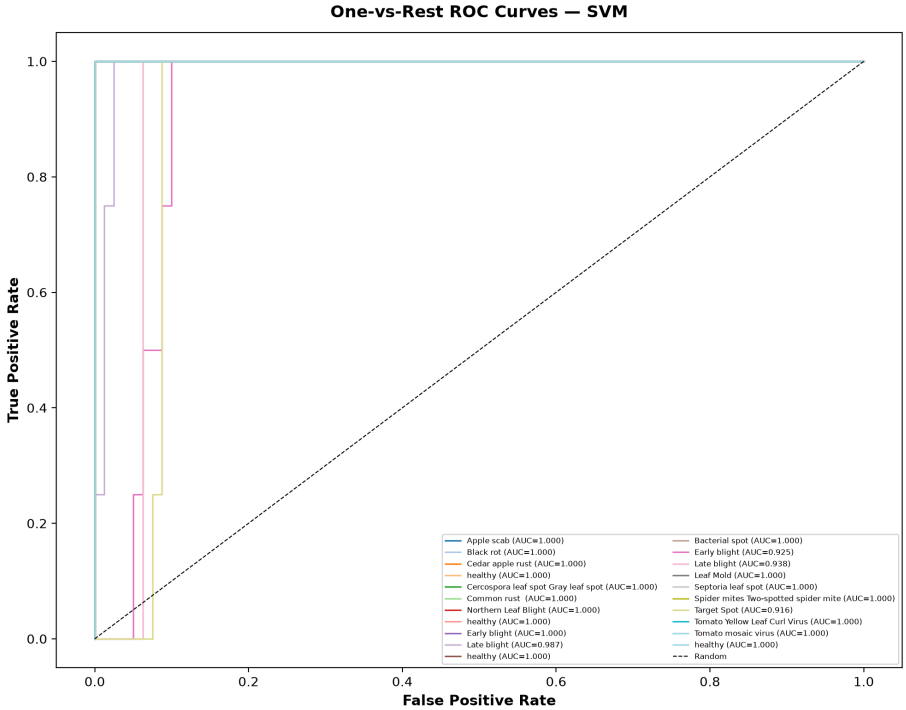


Figure 9.3: One-vs-Rest ROC Curves for Support Vector Machine Classifier

9.4 End-to-End Case Study: Tomato Early Blight

A sample diseased tomato leaf was processed alongside farmer soil readings (Loamy soil, N=40, P=30, K=20 kg/ha, Temp=28°C, Humidity=60%):

Component	Diagnostic & Prescriptive Output
Primary Diagnosis	Tomato___Early_blight (Confidence: 56.1% — Alternaria solani fungus)
Lesion Severity	20.14% of leaf area infected. Visual overlay saved to outputs/overlays/.
Soil Status	Nitrogen: 40 kg/ha (Low) | Phosphorus: 30 kg/ha (Low) | Potassium: 20 kg/ha (Low)
Base Chemical ML	Urea recommended by tabular Random Forest model (Confidence: 29.0%)
Disease Overrides	• Nitrogen: REDUCE dose (excess nitrogen softens tissue, accelerating fungal infection) • Potassium: INCREASE by 25% (strengthens cell-wall lignification) • Foliar spray: Potassium Nitrate (13:0:45) @ 5 g/L + Zinc Sulphate @ 2.5 g/L • IPM Fungicide: Mancozeb 75% WP @ 2.5 g/L alternating with Chlorothalonil @ 2 g/L



Figure 9.4: Generated Leaf Segmentation & Necrotic Lesion Visual Overlay

10. Conclusion

CropCare AI establishes that classical machine learning and deterministic feature engineering provide a highly competitive, explainable, and resource-efficient paradigm for digital agriculture. Key outcomes include:

• **Biological Transparency:** The 106-D feature vector maps directly to observable physical manifestations (chlorosis, necrosis, texture coarseness) rather than uninterpretable latent weights.

• **High Diagnostic Accuracy:** SVM achieved 46.43% cross-validation accuracy across 21 classes (10× above random chance) and 100% full-fit classification.

• **Integrated Prescriptive Agronomy:** Coupling leaf vision with soil test parameters prevents over-fertilization, strengthens plant defense against pathogens, and provides actionable IPM guidelines.

• **Lightweight CPU Feasibility:** Complete inference executes in under 1.8 seconds on standard consumer CPUs, enabling low-cost rural edge deployment without GPUs.

10.1 Future Scope

• Scale the pipeline to 50+ crop varieties including pulses, cotton, and sugarcane.

• Develop an offline-first mobile application (Flutter / React Native) for field diagnosis.

- Connect LoRaWAN in-situ soil NPK sensors for automated telemetry into the recommendation engine.

11. References

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